Module 7: Building AI Systems

Inputs, generation, tools, evaluation, and artifacts

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Primary sources: Osmani, Chapter 7; Thomas & Hunt, Tracer Bullets review.

Learning Outcomes

- Analyze components of an AI-assisted system.
- Describe data and context flow.
- Evaluate integration challenges.
- Design a basic architecture for an AI-assisted feature.

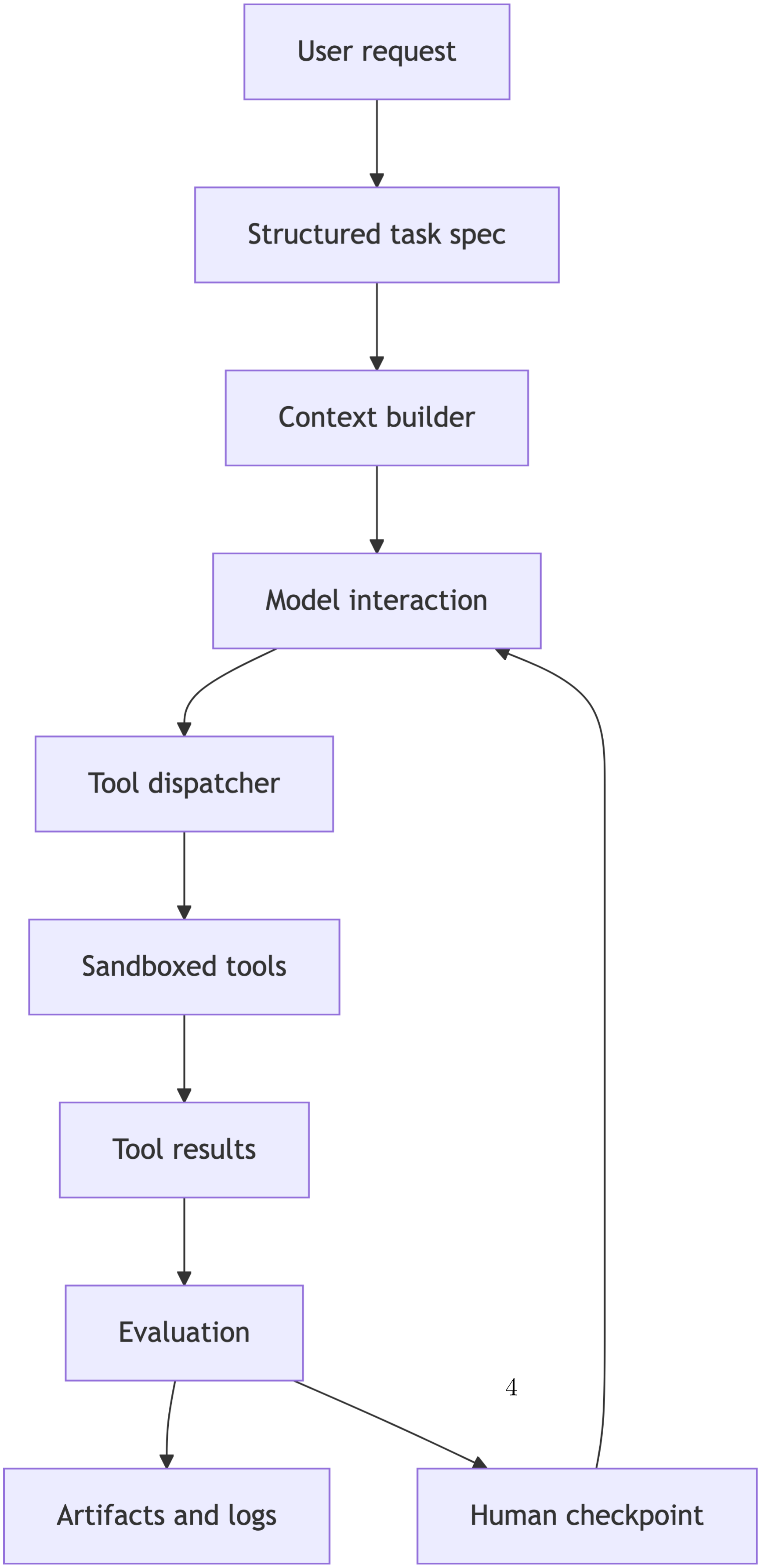
From Task to System

An AI-assisted system is more than a model call.

It includes:

- Input handling.
- Context assembly.
- Prompt construction.
- Model interaction.
- Tool invocation.
- Evaluation.
- Artifact tracking.
- Human oversight.

Controlled Agent Architecture



System Boundary

Define what the agent may and may not do.

For a controlled coding agent:

- Allowed tools.
- Allowed paths.
- Allowed commands.
- Required approvals.
- Required logs.
- Required tests.

Context Flow

Context should be curated, not dumped.

Typical flow:

1. User request.
2. Structured task specification.
3. Relevant project files or summaries.
4. Tool definitions.
5. Evaluation criteria.
6. Output and artifact records.

Integration Challenges

- Mismatched frontend/backend contracts.
- Unclear data models.
- Generated code that bypasses abstractions.
- Tests that require unavailable services.
- Tool outputs too verbose to inspect.
- Error handling split across layers.

Tracer Bullet Architecture

Build one thin complete workflow first.

Example:

- Read task.
- Ask model for plan.
- Dispatch one safe tool.
- Capture result.
- Evaluate with one check.
- Save run artifact.

AI System Prompting

Useful prompts ask the model to operate inside architecture.

- "Use only these tools."
- "Return JSON matching this schema."
- "If input is ambiguous, ask for clarification."
- "Do not modify files outside this sandbox."
- "Explain verification steps separately from implementation."

Lab 4 Development Connection

Lab 4 begins risk analysis and production readiness.

Your risk analysis should be specific to your actual system:

- Tool misuse.
- Sandbox escape.
- Prompt injection.
- False confidence from weak tests.
- Missing human checkpoint.

Practice Activity

Draw your agent as boxes and arrows.

For each arrow, label:

- Data passed.
- Trust boundary.
- Validation step.
- Failure mode.

Takeaways

- AI engineering is systems engineering.
- Context, tools, and evaluation are first-class components.
- Thin end-to-end slices make risk visible.